

BETHEL, AK, USA

WMO: 702190

Lat: 60.785N

Lon: 161.829W

Elev: 31

StdP: 100.95

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26615

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.1	-29.3	-34.4	0.1	-31.1	-32.1	0.2	-28.7	16.1	-10.3	14.7	-13.5	4.1	20	0.833

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.4	22.8	15.1	20.8	14.1	18.8	13.4	16.1	20.6	15.1	19.1	14.2	17.7	4.5	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.6	10.4	17.2	13.5	9.7	16.0	12.4	9.0	15.0	45.1	20.5	42.1	19.2	39.7	17.7	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.0	11.7	10.5	DB	-33.7	26.5	3.3	3.0	-36.1	28.7	-38.0	30.5	-39.8	32.2	-42.2	34.3	(4)
			WB	-32.9	17.6	2.5	1.8	-34.6	18.9	-36.1	20.0	-37.5	21.0	-39.3	22.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.2	-13.6	-9.9	-9.8	-1.8	6.0	11.8	13.4	12.1	8.0	0.1	-7.6	-12.2	(6)
(7)		DBStd	11.62	9.77	8.57	7.61	6.08	4.30	3.02	2.85	2.60	3.20	5.07	7.10	8.72	(7)
(8)		HDD10.0	4012	732	556	613	354	134	14	4	7	74	308	527	688	(8)
(9)		HDD18.3	6784	990	789	872	604	382	197	157	193	311	566	777	946	(9)
(10)		CDD10.0	272	0	0	0	0	11	68	107	73	13	0	0	0	(10)
(11)		CDD18.3	4	0	0	0	0	0	1	2	1	0	0	0	0	(11)
(12)		CDH23.3	56	0	0	0	0	1	19	29	8	0	0	0	0	(12)
(13)		CDH26.7	11	0	0	0	0	0	2	8	1	0	0	0	0	(13)
(14)	Wind	WSAvg	5.1	6.2	6.0	5.6	5.1	4.7	4.3	4.6	4.5	4.9	5.3	5.6	(14)	
(15)	Precipitation	PrecAvg	490	21	19	19	19	27	44	67	80	74	44	45	30	(15)
(16)		PrecMax	605	65	76	88	57	67	107	104	165	176	112	107	90	(16)
(17)		PrecMin	300	2	2	2	2	5	9	18	26	21	4	2	8	(17)
(18)		PrecStd	78	17	17	17	14	17	19	25	30	37	26	26	18	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.7	4.4	5.7	13.2	21.4	25.9	26.8	24.0	18.1	12.0	6.9	4.9	(19)
(20)			MCWB	2.7	2.3	3.1	7.1	11.9	15.4	17.5	16.6	12.3	8.6	5.3	2.6	(20)
(21)		2%	DB	3.1	3.0	3.2	10.5	18.1	22.6	23.0	20.6	15.6	9.9	4.5	3.0	(21)
(22)			MCWB	1.5	1.2	1.4	5.7	10.8	14.2	15.7	15.0	11.5	7.7	3.5	1.6	(22)
(23)		5%	DB	1.9	2.2	2.1	8.1	15.4	20.2	20.9	18.5	14.0	8.1	3.0	1.8	(23)
(24)			MCWB	0.8	0.9	0.8	4.6	9.5	13.2	14.7	14.0	10.8	6.6	2.2	0.9	(24)
(25)		10%	DB	0.3	1.1	0.9	6.0	13.3	18.0	18.7	16.7	12.5	6.8	1.7	0.4	(25)
(26)			MCWB	-0.3	0.3	0.0	3.4	8.3	12.3	14.0	13.4	10.2	5.6	1.1	-0.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.9	2.7	3.4	7.7	12.9	16.7	19.2	17.8	13.6	9.6	5.6	3.0	(27)
(28)			MCDWB	4.4	3.9	5.3	12.3	19.2	23.5	25.7	21.9	16.1	11.0	6.4	4.3	(28)
(29)		2%	WB	1.7	1.7	1.7	6.0	11.3	15.0	16.4	15.9	12.2	8.0	3.7	1.8	(29)
(30)			MCDWB	2.8	2.6	2.9	9.9	17.0	20.7	21.2	19.1	14.5	9.5	4.4	2.7	(30)
(31)		5%	WB	0.8	1.0	0.8	4.8	10.0	13.9	15.3	14.8	11.3	6.9	2.3	0.9	(31)
(32)			MCDWB	1.7	1.9	1.8	8.0	15.0	19.3	19.8	17.3	13.2	8.1	3.0	1.7	(32)
(33)		10%	WB	-0.3	0.3	0.0	3.5	8.7	12.8	14.4	13.8	10.5	5.7	1.0	-0.2	(33)
(34)			MCDWB	0.4	1.1	1.0	5.9	12.9	17.2	18.1	16.2	12.2	6.7	1.6	0.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.9	6.9	7.8	8.3	9.1	9.0	7.4	6.6	6.8	6.0	5.9	6.4	(35)
(36)			MCDBR	6.5	6.3	7.3	10.9	13.1	13.3	11.8	9.9	8.8	6.7	5.2	6.0	(36)
(37)			MCWBR	5.7	5.4	6.0	7.0	6.9	5.9	5.2	4.9	5.1	4.9	4.6	5.3	(37)
(38)		5% WB	MCDWB	6.2	6.1	6.9	10.3	12.6	11.8	10.5	8.4	7.3	6.1	5.1	6.1	(38)
(39)			MCWBR	5.6	5.4	5.8	6.7	6.8	5.5	5.0	4.7	4.7	4.8	4.7	5.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.227	0.285	0.289	0.354	0.376	0.365	0.373	0.381	0.341	0.295	0.248	0.215	(40)	
(41)		taud	2.086	2.156	2.228	2.145	2.293	2.405	2.418	2.382	2.455	2.420	2.073	1.927	(41)	
(42)		Ebn at Noon	525	693	836	832	845	864	846	805	779	699	467	335	(42)	
(43)		Edh at Noon	40	71	92	123	115	105	102	98	76	55	40	29	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.37	1.07	2.91	4.49	5.06	5.20	4.29	3.36	2.37	1.22	0.43	0.19	(44)	
(45)		RadStd	0.06	0.14	0.31	0.38	0.47	0.45	0.42	0.39	0.27	0.09	0.06	0.02	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (7 neighbors)	N/A	+1.67	+1.60	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page